

Matthew Wong

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Education

University of California, Irvine - GPA: 3.53

Irvine, CA

B.S. Computer Science and Business Information Management

Expected June 2028

- **Relevant Coursework:** Data Structures, Business Data Management, Machine Learning and Data Mining, Exploratory Data Analysis, Linear Algebra

Experience

Undergraduate Research Assistant

April 2026 – Present

UC Irvine - Prof. John Joseph

Irvine, CA

- Built a Python NLP pipeline using TF-IDF and logistic regression to classify 600+ research papers, improving accuracy from 71% (abstract-only baseline) to 87%; analyzed model performance and documented methods
- Supported RAG financial-forecasting research through preparing and validating datasets with Python

Undergraduate Research Intern

March 2026 – August 2026

UC Irvine - Prof. Patrick Bergemann

Irvine, CA

- Built Python web-scraping and OCR data pipelines to ingest unstructured sources and produce clean datasets
- Owned data quality for 200,000+ athlete records with pandas and regex by enforcing schemas and validation rules
- Modeled longitudinal athlete datasets and performed statistical analysis to generate insights on athlete patterns

Global Business Brigades @ UCI

March 2025 – May 2025

Strategy Consultant

Irvine, CA

- Partnered with client stakeholders to gather data requirements, analyze business operations, and prioritize data-informed process improvements
- Created a centralized information hub and presented actionable recommendations to client leadership

Projects

Ginseng Museum Data Platform | *Python, PostgreSQL, Snowflake, dbt, Airflow, Power BI*

September 2026

- Building a PostgreSQL data model and SQL transformation workflows to integrate records across customer data
- Developing an Airflow ELT pipeline with incremental loading, duplicate detection, validation checks, and logging
- Creating dbt models and Power BI dashboards to surface revenue, refund, product, and customer KPIs for downstream business analysis

PitchVision: Soccer Transition Analytics | *Python, OpenCV, YOLO*

August 2026

- Developed a computer-vision pipeline with YOLO and OpenCV to track players and analyze team behavior surrounding possession changes in match footage
- Quantifying transition moments through possession duration, ball-recovery time, player movement, and immediate post-turnover actions to compare successful and unsuccessful transitions

DatathonPD: Crime Data Anomaly Detection | *Python, SQL, scikit-learn, pandas, NumPy*

April 2026

- Built an end-to-end Python and SQL data pipeline for 1M+ crime records, resolving invalid values and engineering reporting-delay features
- Implemented and evaluated Local Outlier Factor and Isolation Forest, flagging 3% of records with anomalous reporting behavior and documenting data-quality implications

TECHNICAL SKILLS

Programming & ML: Python, SQL, R, scikit-learn, pandas, NumPy

AI/ML & NLP: RAG, NLP, TF-IDF, OCR, OpenCV, YOLO

Data & Cloud: PostgreSQL, Snowflake, dbt, Apache Airflow, ETL/ELT

Development & Tools: Git, Docker, Streamlit, Power BI, Jupyter Notebooks, Microsoft Excel